

Question:

Tom's age is \$T\$ years, which is also the sum of the ages of his three children. His age \$N\$ years ago was twice the sum of their ages then. What is \$T/N\$?

\$(A) \ 2 \quad (B) \ 3 \quad (C) \ 4 \quad (D) \ 5 \quad (E) \ 6\$

Reasoning + Answer:

<think> To solve this problem, we need to set up a system of equations based on the given information. Let the ages of Tom's three children be (a) , (b) , and (c) years. According to the problem, Tom's age (T) is equal to the sum of the ages of his three children, so we have:

$$T = a + b + c.$$

N years ago, Tom's age was $(T - N)$ and the ages of his children were $(a - N)$, $(b - N)$, and $(c - N)$. The problem states that (N) years ago, Tom's age was twice the sum of the ages of his children at that time. Therefore, we can write the equation:

$$T - N = 2[(a - N) + (b - N) + (c - N)].$$

Substituting $(T = a + b + c)$ into the equation, we get:

$$a + b + c - N = 2[(a - N) + (b - N) + (c - N)].$$

Simplifying the right side of the equation:

$$a + b + c - N = 2(a + b + c - 3N).$$

Expanding and simplifying further:

$$a + b + c - N = 2a + 2b + 2c - 6N.$$

Rearranging terms to isolate terms involving $(T = a + b + c)$:

$$a + b + c - N - 2a - 2b - 2c + 6N = 0,$$
$$-a - b - c + 5N = 0,$$
$$a + b + c = 5N,$$
$$T = 5N.$$

Therefore, the ratio (T/N) is:

$$\frac{T}{N} = \frac{5N}{N} = 5.$$

<answer>5</answer>